

SANA: LakeQA SUBSET20B results

Experiment records 2026-08-31-web-arm-subset20b and 2026-08-31-model-tiers-subset20b

1 Results

1.1 Retrieval source: open web versus the data lake

To test whether the retrieval axis reflects the data lake specifically or a more general property of the agent, we replaced lake search with open-web search. The arm uses the Parallel Search API and, under `--no-s3`, loses every lake-backed file tool: the agent keeps only `download`, which fetches an arbitrary `http(s)` URL into its sandbox, and `execute_code`. All arms hold the remaining axes at `search_results=naive`, `profile=standard` and `computation_tool=standard`, since the ideal computation tools resolve against authored data-lake records and would make a web run’s outcome independent of what it retrieved. Each arm was run three times.

The web arm reaches 38.3% against 56.7% for the strongest lake arm, a gap of 18.3pp that clears the ± 12 pp two-sigma replicate threshold measured across these arms.

The more informative quantity is where retrieval breaks down. D_{ret} is the fraction of gold datasets that appeared in search results; D_{acc} is the fraction the agent actually read. All three lake arms surface gold at a similar rate (Ideal 82.5%, Standard 84.1%, Naive 85.6%), so they are not separated by what search returns. They are separated by the drop between finding and reading (Ideal 3.4pp, Standard 16.2pp, Naive 15.7pp): oracle search wins because the agent acts on what it was shown, not because it was shown more.

For the web arm both quantities are undefined rather than zero: it touches no lake datasets by construction, so it can neither retrieve nor read a gold source.

Table 1: Retrieval-source comparison on LakeQA SUBSET20B (gpt-5-mini, 20 tasks per arm, three replicate rounds, semantic match). $\bar{x}$ is the mean over the three rounds, *Rounds/task* the mean number of agent cycles a task took, and cost the mean spend per task over completed rows, including any hidden helper agents. D_{ret} and D_{acc} are retrieval and access recall of gold datasets, undefined for the web arm.

Arm	$\bar{x}$ (%)	Rounds/task	\$/task	D_{ret} (%)	D_{acc} (%)
Ideal	56.7	23.1	0.0406	82.5	79.1
Standard	51.7	27.6	0.0426	84.1	67.9
Naive	56.7	27.2	0.0432	85.6	69.9
Web	38.3	25.4	0.0502	n/a	n/a

What the web arm actually measures. `download` accepts any URL, and the model routinely constructs one from pretrained knowledge rather than following a search result: live Socrata queries against `data.cityofnewyork.us`, `data.va.gov` export endpoints, and a timestamped `web.archive.org` URL. The arm therefore measures an agent with open web access and code execution, not web-search retrieval, and its data path may be stronger than the lake arms’ since it aggregates live authoritative sources rather than a fixed snapshot. That liveness also produces false negatives: two failures were arithmetically correct against current data but disagreed with gold answers derived from the snapshot (2,700,968 against a gold 2,705,988; 2923.59 against 2941).

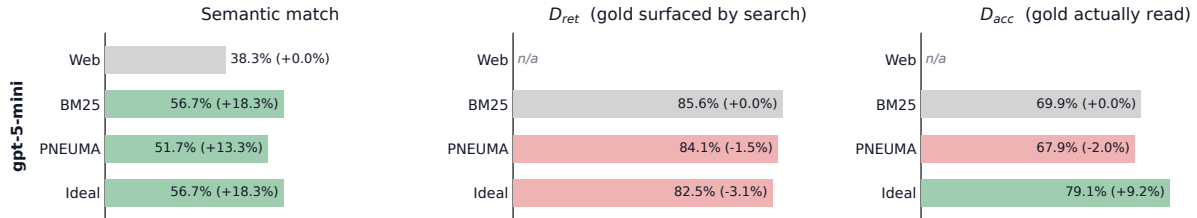


Figure 1: Retrieval source as an axis. Arms are ordered by how much of the lake they can see, so the baseline row is the open web and each delta is what moving into the lake buys. The right two panels are baselined on BM25, the weakest arm with a defined recall. Reading them together is the point: D_{ret} is flat across the three lake arms – oracle search surfaces slightly *less* gold than BM25 – while D_{acc} rises with accuracy. What separates the arms is not what search returns but how much of it the agent reads. The web arm touches no lake dataset by construction, so its recall is undefined rather than zero.

1.2 Replicate variance on the ablation grid

We ran the leave-one-out ablation over four model tiers on SUBSET20B, a 20-task split of LakeQA stratified jointly on node count and historical solve rate. Every axis is held at the oracle and one axis is degraded per cell, giving seven cells per model. Each cell was run **three times** under identical configuration, so every number carries a standard deviation rather than resting on a single sample.

That replication is the central result. The mean per-cell standard deviation is 6.4 pp, so a two-sigma interval spans ± 13 pp and an effect must exceed roughly 13 pp to be distinguishable from run-to-run variance. Since one task is worth 5 pp at $n = 20$, that threshold is about 3 tasks.

Of the 24 degraded cells, only 5 clear that threshold: Search: BM25 at **gpt-5.4-nano** (-25.0 pp), Plan: Default at **gpt-5.2** (-20.0 pp), Search: BM25 at **gpt-5-mini** (-16.7 pp), Search: BM25 at **gpt-5.2** (-15.0 pp), Search: PNEUMA at **gpt-5-mini** (-15.0 pp). 4 of those are search degradations, on 3 of the 4 tiers; the remaining cells cannot be separated from noise at this sample size. Search costs the most accuracy of any axis, but it clears the threshold on the weaker tiers and not on the strongest – the same degradation hurts less as the model improves.

The reference cells themselves are **gpt-5.4-nano** 41.7% ($\sigma = 2.9$), **gpt-5-mini** 73.3% ($\sigma = 2.9$), **gpt-5.2** 71.7% ($\sigma = 10.4$), **gpt-5.6-luna** 75.0% ($\sigma = 5.0$). The two strongest, **gpt-5.6-luna** and **gpt-5-mini**, differ by 1.7 pp against a standard error of 3.3 pp on that difference, so they are not separable here.

Accuracy is not what separates the tiers. Over all cells the top three land within 7.9 pp of one another (64.0–71.9%), inside the replicate threshold, while cost per correct answer spans 13 $\times$: **gpt-5.6-luna** answers at \$0.0207 in 10.1 agent cycles against **gpt-5.2** at \$0.2760 in 16.3. The tier that costs the most buys nothing here that the cheapest does not already provide.

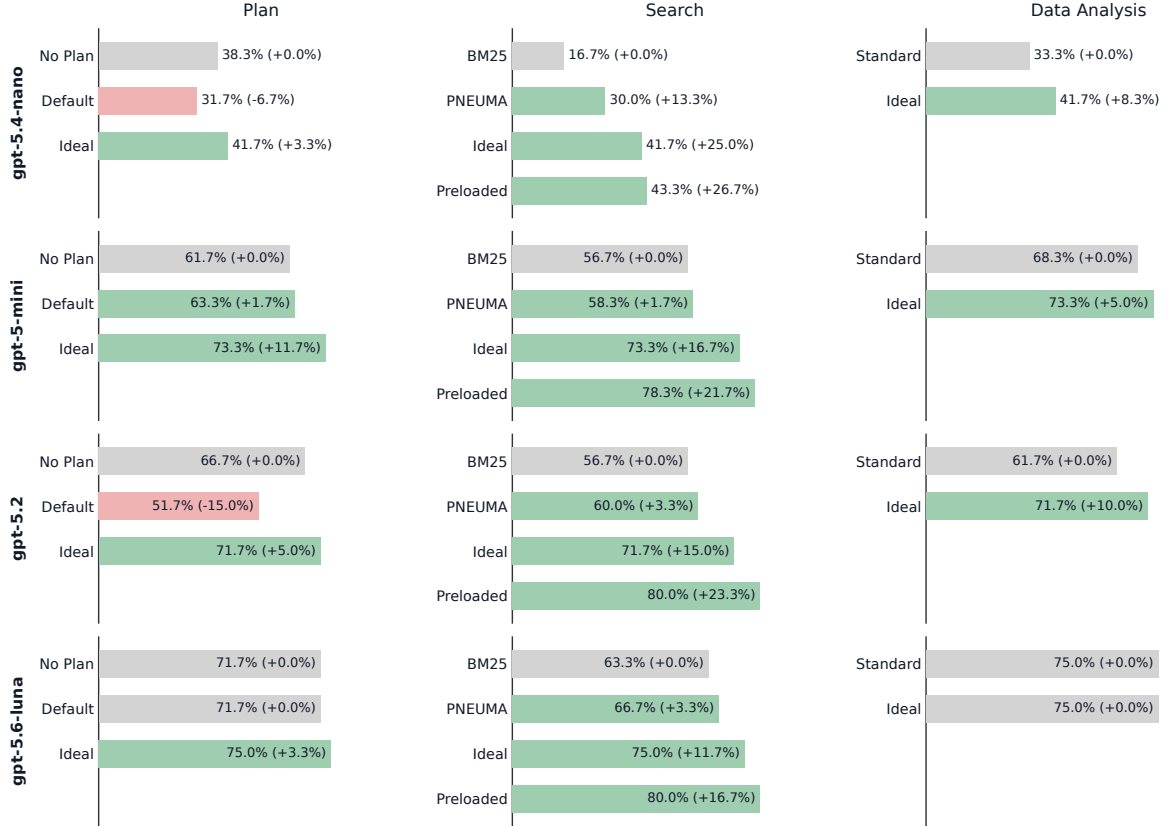


Figure 2: Per-axis ablation across model tiers on LakeQA SUBSET20B. Rows are models, columns are the three SANA axes, and each bar is one mode of that axis with the other two held at *Ideal*. The label gives the semantic match and its change against the panel’s baseline – the weakest mode of that axis, the top row of each panel. Bars are means over the replicate rounds; colour repeats the sign of the printed delta rather than carrying it alone. Against the ± 13 pp two-sigma replicate threshold of Table 3, only the Search panel moves any model further than run-to-run variance.

Table 2: Leave-one-out ablation across model tiers on LakeQA SUBSET20B (20 tasks per cell, semantic match, mean over the replicate rounds each row has, run under identical configuration). $\bar{x}$ is the mean over those rounds and δ the effect relative to that model’s reference cell; effects exceeding the ± 13 pp two-sigma threshold of Table 3 are set in bold. *Rounds/task* is the mean number of agent cycles a task took. Cost is the mean spend per task over completed rows, including the hidden helper agents the ideal modes delegate to – these outweigh the visible agent, so a cost counted on the main agent alone would rank the oracle tools as the cheap ones.

Model	Condition	$\bar{x}$ (%)	δ (pp)	Rounds/task	\$/task
gpt-5.4-nano	Reference (all ideal)	41.7	—	20.5	0.0351
	Search: BM25	16.7	−25.0	21.0	0.0345
	Search: PNEUMA	30.0	−11.7	20.9	0.0364
	Search: Preloaded	43.3	+1.7	20.0	0.0391
	Plan: No Plan	38.3	−3.3	18.9	0.0348
	Plan: Default	31.7	−10.0	22.2	0.0369
	Data An.: Standard	33.3	−8.3	21.9	0.0177
gpt-5-mini	Reference (all ideal)	73.3	—	23.1	0.0781
	Search: BM25	56.7	−16.7	29.0	0.0927
	Search: PNEUMA	58.3	−15.0	27.2	0.0987
	Search: Preloaded	78.3	+5.0	16.7	0.0595
	Plan: No Plan	61.7	−11.7	21.3	0.0784
	Plan: Default	63.3	−10.0	23.7	0.0817
	Data An.: Standard	68.3	−5.0	22.4	0.0399
gpt-5.2	Reference (all ideal)	71.7	—	17.3	0.1916
	Search: BM25	56.7	−15.0	19.7	0.2275
	Search: PNEUMA	60.0	−11.7	19.0	0.2251
	Search: Preloaded	80.0	+8.3	12.3	0.1420
	Plan: No Plan	66.7	−5.0	12.9	0.1615
	Plan: Default	51.7	−20.0	16.1	0.1701
	Data An.: Standard	61.7	−10.0	16.7	0.1194
gpt-5.6-luna	Reference (all ideal)	75.0	—	10.4	0.0148
	Search: BM25	63.3	−11.7	11.9	0.0200
	Search: PNEUMA	66.7	−8.3	12.7	0.0193
	Search: Preloaded	80.0	+5.0	8.2	0.0134
	Plan: No Plan	71.7	−3.3	7.9	0.0128
	Plan: Default	71.7	−3.3	10.1	0.0142
	Data An.: Standard	75.0	+0.0	9.9	0.0096

Table 3: Replicate noise floor, over every cell run three times. One task is worth 5 pp at $n = 20$, so the mean per-cell standard deviation is close to a single task and the two-sigma threshold is about 3.

Statistic over per-cell σ	Value (pp)
Mean	6.4
Median	6.7
Maximum	14.4
Two-sigma threshold	12.8
Cells	28